

F1 TELEMETRY ANALYSIS

Abu Dhabi 2021 — Qualifying Session

Hamilton vs Verstappen · The Lap That Shaped a Championship

	HAM Lewis Hamilton	VER Max Verstappen
Lap Time	1:22.242	1:21.990
Top Speed	320.9 km/h	325.0 km/h
Avg Throttle	84.6 %	82.1 %
Time Braking	11.7 %	12.4 %
Gap	+0.252 s	POLE POSITION

This document accompanies the open-source F1 Telemetry Dashboard — a Python project that pulls official race data via FastF1 and visualizes it interactively with Dash and Plotly. All data is sourced from the official F1 timing feed.

1. Context — Why This Lap Matters

The 2021 Abu Dhabi Grand Prix weekend was the final round of one of the most contested Formula 1 seasons in history. Heading into that weekend, Lewis Hamilton (Mercedes) and Max Verstappen (Red Bull) were level on championship points — the first time in modern F1 that the title went to the final race with a dead heat in the standings.

Qualifying mattered enormously. Starting from pole at Yas Marina's long straights is a significant strategic advantage — and Verstappen took it, setting the fastest lap in Q3 by 0.252 seconds. Understanding *where* that gap was built, and *how*, is exactly what telemetry analysis is for.

The Yas Marina Circuit in Abu Dhabi features three long straights where top speed and drag level have an outsized impact on lap time. In 2021, the Red Bull RB16B was known for its low-drag philosophy — a setup that paid dividends on circuits like this.

2. Methodology

2.1 Data Source

All telemetry was retrieved from the official F1 timing feed using the **FastF1** Python library (v3.x). FastF1 provides access to car telemetry at approximately 240 Hz — position, speed, throttle, brake, DRS, gear, and more — as well as lap timing and session metadata.

For this analysis we extracted the single fastest qualifying lap for each driver from the Q3 session:

- HAM — 1:22.242 (Q3 fastest lap)
- VER — 1:21.990 (Q3 fastest lap, pole position)

2.2 Interpolation and Alignment

A fundamental challenge in comparing two laps is that their telemetry samples are not aligned in time — each driver's dataset has a different number of samples and different time offsets. To enable a fair point-by-point comparison, the raw telemetry was processed as follows:

1. The raw distance column was **normalized** to [0, 1], where 0 and 1 both represent the start/finish line.
2. Each channel (Speed, Throttle, Brake, Time) was **interpolated** using NumPy's 1-D linear interpolator onto a shared grid of 1,000 equally-spaced distance points.
3. Time was converted to **elapsed seconds from lap start**, enabling the time delta chart (who was ahead and by how much at each point).

2.3 Time Delta Calculation

The delta channel is computed as:

```
delta[i] = VER_time_from_start[i] - HAM_time_from_start[i]
```

A positive value means VER took longer to reach distance point *i*, i.e., Hamilton is leading at that point. A negative value means Verstappen is ahead.

3. Key Findings

3.1 Straight-Line Speed

Verstappen's top speed of **325.0 km/h** was **4.1 km/h** higher than Hamilton's 320.9 km/h. This gap reflects the fundamental difference in aerodynamic philosophy between the two cars in 2021. Red Bull ran a lower downforce, lower drag setup that sacrificed some cornering grip for top speed on the straights — a trade-off that favoured Yas Marina's long DRS zones.

Note: Hamilton was running marginally higher average throttle (84.6% vs 82.1%), suggesting the Mercedes driver was pushing harder in the traction phases to compensate for a slight deficit in raw straight-line pace.

3.2 Braking Behaviour

Verstappen spent a slightly higher proportion of the lap on the brakes (12.4% vs 11.7% for Hamilton). This is consistent with a low-downforce car that relies more on late braking to compensate for lower corner speeds — a hallmark of the Red Bull driving style that both Verstappen and his engineers optimized for.

3.3 Time Delta — Who Was Leading, and When

The time delta tells the most complete story of how the lap evolved:

Phase	Distance	Who Led	Gap
Sector 1 entry	0.00 – 0.10	HAM	+0.091 s
Mid-lap (S2/S3)	~0.50	VER	–0.309 s (max)
Final crossing	1.00	VER	–0.161 s
Full lap result	—	VER POLE	–0.252 s

Hamilton briefly had a marginal advantage in the very first portion of the lap — corresponding to the tighter Turn 1–5 complex where the Mercedes' higher downforce setup provided slightly more grip. However, as soon as the cars hit the long back straight (around distance 0.35–0.55), Verstappen's lower drag began to compound, and he built his maximum advantage of 0.309 seconds at the midpoint.

Hamilton clawed some time back in the final sector, but the deficit was already too large. Verstappen crossed the line 0.252 seconds ahead — pole secured.

Key takeaway: the Red Bull's advantage was almost entirely structural — it was built by the car's aerodynamic package, not by a dramatic difference in driving input. Both drivers were pushing at the limit. The chassis made the difference.

4. Technology Stack

The project is entirely Python-based and open-source. Every tool used is free.

Library	Version	Role
FastF1	3.x	Official F1 telemetry data retrieval and session management
Pandas	2.x	Data loading, manipulation, filtering
NumPy	1.x	Numerical interpolation onto shared distance grid
Dash	2.x	Web-based interactive dashboard framework
Plotly	5.x	Interactive charts (speed, throttle, brake, delta)
Python	3.10+	Runtime environment

Dashboard Views

The interactive dashboard (`dashboard.py`) includes five views, selectable with a radio button:

View	What You See
Speed	Speed trace (km/h) of both drivers across the full lap
Throttle	Throttle position (%) — shows aggressive vs smooth application
Brake	Brake trace — when and how long each driver was on the brakes
Time Delta	Cumulative time gap (VER – HAM) showing who led at each point
Full View	All four charts stacked — default view on launch

5. Project Structure

```
01_TELEMETRY_F1/
├── data/
│   ├── abu_dhabi_2021_ham.csv ← HAM raw telemetry
│   ├── abu_dhabi_2021_ver.csv ← VER raw telemetry
│   └── abu_dhabi_2021_comparison.csv ← Aligned dataset (1000 pts)
├── notebooks/
│   ├── 01_visualizacion_validacion.ipynb
│   └── 02_telemetry_preprocessing.ipynb
├── src/
│   ├── prepare_data_set.py ← Downloads & preprocesses via FastF1
│   └── dashboard.py ← Launches the interactive Dash app
```

■■■ requirements.txt

The pre-processed CSV files are included so the dashboard can be run immediately without needing to download data. To regenerate from scratch (e.g., for a different race or session), run `prepare_data_set.py` with FastF1 installed.

6. What's Next in This Portfolio

This is the first entry in an ongoing F1 Data Analysis portfolio. Planned projects:

- **02 – Suspension Analysis:** CAD and data analysis of F1 suspension geometry
- **03 – Tire Degradation:** Compound behavior across race stints
- **04 – Sector Breakdown:** Mini-sector analysis across multiple drivers
- **05 – 2024 Season:** Updated telemetry comparisons with current-generation cars

Each project includes full source code, pre-processed data, and a Jupyter notebook walkthrough. Supporting the portfolio helps fund the time needed to research, build, and document each analysis properly.

F1 Data Portfolio • Built with FastF1, Pandas, Dash & Plotly • Open Source